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Studierichting: Informatica
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$$\int \frac{\tan x dx}{1 + \cos^2 x}$$

$$t = \tan x$$

$$dt = (1 + t^2)dx$$

$$\frac{dt}{1 + t^2} = dx$$

$$\Rightarrow \int \frac{t}{1 + \frac{1}{1 + t^2}} \cdot \frac{1}{1 + t^2} dt$$

$$= \int \frac{t}{\frac{2 + t^2}{1 + t^2}} \cdot \frac{1}{1 + t^2} dt$$

$$= \int \frac{t}{2 + t^2} dt$$

$$u = 2 + t^2$$

$$du = 2t dt$$

$$\frac{du}{2} = t dt$$

$$\Rightarrow \frac{1}{2} \int \frac{1}{u} du$$

$$= \frac{\ln |2 + \tan^2 x|}{2} + C$$

References